

Dashboard 6: Borrower Behavior & Loan Performance Analysis

Introduction:

Analyzing loan profitability, interest earnings, recovery trends, and risk exposure is essential for maintaining a resilient financial portfolio. By focusing on these metrics, the dashboard delivers clear insights into revenue generation, repayment stability, and financial risks while incorporating key factors such as **borrower behavior, loan performance trends, repayment patterns, loan repayment, delinquency, borrower demographics, and financial behavior**. These insights empower stakeholders to identify high-value segments, mitigate risks, and optimize lending strategies for stronger portfolio performance.

DAX Queries:

1. Average Loan Repayment Period (Months):

```
Average Loan Repayment Period (Months) = AVERAGE([Repayment  
Period (Months)])
```

2. Late Payment Percentage (%):

```
Late Payment (%) =  
DIVIDE (  
    CALCULATE (  
        COUNTROWS (bondora_data),  
        bondora_data[status] = "Late"  
    ),  
    COUNTROWS (bondora_data),  
    0  
)
```

3. Borrower Income vs Loan Amount Ratio:

```
Income-to-Loan Ratio =  
DIVIDE (  
    AVERAGE (bondora_data[incometotal]),  
    AVERAGE (bondora_data[amount]),  
    0  
)
```

4. Loan Default Rate by Borrower Age Group (%):

```
Loan Default Rate by Age Group (%) =  
VAR TotalLoans =  
    CALCULATE (COUNTROWS ('bondora_data'))  
  
VAR DefaultedLoans =  
    CALCULATE (  
        COUNTROWS ('bondora_data'),  
        'bondora_data'[Status] = "Late"  
    )  
RETURN  
DIVIDE (DefaultedLoans, TotalLoans)
```

5. Early Repayment Percentage (%):

```
Early Payment (%) =  
DIVIDE (  
    CALCULATE (  
        COUNTROWS (bondora_data),  
        bondora_data[status] = "Current"  
    ),  
    COUNTROWS (bondora_data),  
    0  
)
```

6. Borrower Dependence on High-Interest Loans:

```
High-Interest Loan Dependence =  
VAR ThresholdPrincipal = 5000    -- set your threshold for  
high principal balance  
VAR HighPrincipalLoans =  
    CALCULATE (  
        COUNTROWS( bondora_data ),  
        bondora_data[principalbalance] > ThresholdPrincipal  
    )  
VAR TotalLoans =  
    COUNTROWS (bondora_data )  
RETURN  
DIVIDE( HighPrincipalLoans, TotalLoans )
```

Visualizations:

Visualization	Visualization Type
Avg.Loan Repayment (Months)	KPI Card
Late Payment (%)	KPI Card
Income-to-Loan Ratio	KPI Card
Age-wise Default Rate (%)	KPI Card
Early Payment (%)	KPI Card
Dependence on High-Interest Loans	KPI Card
Borrower Age vs Loan Default Rate Bar Chart	X-axis: Default Rate (%) Y-axis: Age_Group
Income-to-Loan Ratio Scattered Plot	X-axis: SUM (amount) Y-axis: SUM (incometotal) Values: Rating
Loan Repayment Behavior Line Chart	X-axis: loanapplicationstartddate Y-axis: COUNT (Status)
Early Repayment Trend Histogram	X-axis: Repayment Period (Months) [bins] Y-axis: COUNT(Status)
High-Interest Loan Dependence Pie Chart	Values: COUNT (loanid) Legend: APR_Depedency_Segment

Filters & Slicers:

- Loan Amount Range
- Borrower Age Group
- Loan Status
- Income Range
- Loan Origination Year
- Last Payment Year

Conclusion:

Identifying borrowers at **higher risk of default** by examining key demographic and financial factors—such as **age**, **income levels**, and **loan terms**—allows lenders to perform a more precise and reliable credit evaluation. By further analyzing **early repayment** patterns, financial institutions can gain a deeper understanding of borrower behavior, enabling them to optimize **loan structuring** and enhance overall **risk assessment** processes. Collectively, these **data-driven insights** not only improve the accuracy of lending decisions but also strengthen **loan recovery strategies** and guide the development of more effective, resilient **credit policies** that support long-term financial stability.